THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · EASY LEVEL UNIT TEST SET 2

Chapters 1–4: **Coordinates, Linear Polynomials, World of Numbers, Algebraic Identities** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Using $(a+b)^2$, the value of 31^2 is: 1

(a) 961 **(b)** 963 **(c)** 960 **(d)** 962

2. The midpoint of the segment joining $(2, 4)$ and $(6, 8)$ is: 1

(a) $(-4, 6)$ **(b)** $(4, 6)$ **(c)** $(4, -6)$ **(d)** $(6, 4)$

3. The slope of the line $y = 3x + 2$ is: 1

(a) 4 **(b)** 5 **(c)** 2 **(d)** 3

4. A decimal expansion that ends after a finite number of digits is called: 1

(a) non-terminating and non-repeating **(b)** irrational
(c) recurring **(d)** terminating

5. Using $a^2 - b^2$, the value of 49×51 is: 1

(a) 2501 **(b)** 2498 **(c)** 2500 **(d)** 2499

6. A point on the x-axis has: 1

(a) ordinate 0 **(b)** equal coordinates
(c) both coordinates 0 **(d)** abscissa 0

7. The graph of a linear polynomial is a: 1

(a) straight line **(b)** pair of points **(c)** parabola **(d)** circle

8. $(-7) \times (-6) =$ 1

(a) 42 **(b)** 44 **(c)** 41 **(d)** 43

9. $x^2 - 25$ factorises as: 1

(a) $(x-5)^2$ **(b)** $(x+5)(x-5)$ **(c)** $(x+5)^2$ **(d)** $(x+25)(x-1)$

10. The reflection of $(3, 4)$ in the x-axis is: 1

(a) $(-3, -4)$ **(b)** $(3, -4)$ **(c)** $(3, 4)$ **(d)** $(-4, 3)$

11. The nth term of the pattern 5, 8, 11, 14, ... is: 1

(a) $3n + 2$ **(b)** $3n + 1$ **(c)** $3n + 3$ **(d)** $4n + 2$

12. $\sqrt{196} =$ 1

(a) 15 **(b)** 16 **(c)** 14 **(d)** 13

13. If $a = 2$ and $b = 5$, then $(a + b)^2 =$ 1

(a) 50 **(b)** 51 **(c)** 49 **(d)** 48

14. The distance between $(1, 3)$ and $(8, 3)$ is: 1

(a) 7 **(b)** 6 **(c)** 8 **(d)** 9

15. The graph of $y = 2x + 4$ cuts the y-axis at: 1

(a) $(1, 4)$ **(b)** $(4, 0)$ **(c)** $(0, -4)$ **(d)** $(0, 4)$

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the zero of the polynomial $3x - 15$. 2

17. Is $\sqrt{16}$ rational or irrational? Give a reason. 2

18. Expand $(x + 8)(x - 8)$. 2

19. Find the midpoint of the segment joining $(2, 10)$ and $(6, 4)$. 2

20. Write the degree of $8x + 3$ and say whether it is constant, linear or quadratic. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Factorise $x^2 + 6x + 9$. 3

22. Find the midpoint of $A(0, 0)$ and $B(2, 4)$ and its distance from the origin. 3

23. Find the nth term and the 10th term of the pattern 3, 7, 11, 15, ... 3

24. Find a rational number between $\frac{1}{4}$ and $\frac{1}{2}$. 3

OR

Evaluate 55×45 using an identity. 3

25. Find the distance between $A(-2, 1)$ and $B(3, 13)$. 3

OR

Find the zero of $3x + 12$ and verify it. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. For the point $P(3, 4)$, state its quadrant, its distances from the two axes and from the origin, and its reflection in the origin. 5

OR

For the line $y = 2x + 6$, write the slope, the y-intercept, the x-intercept, and say whether the point $(3, 12)$ lies on it. 5

27. Divide 5 by 11 and write $\frac{5}{11}$ as a repeating decimal. After how many digits does the block repeat? 5

OR

Factorise $x^2 + 5x + 6$. 5

END OF QUESTION PAPER — Total: 50 Marks

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